Junyi (Conny) Zhou

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Education

- **Harvard University** — Boston, MA
  M.S. in Health Data Science — *August 2025 – March 2027*
  - Program requirements complete December 2026 (commencement March 2027); available for full-time start January 2027
  - Coursework: Deep Learning, AI in Medicine, Healthcare Machine Learning, Data Science
- **Emory University, College of Arts and Sciences** — Atlanta, GA
  B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 — *August 2021 – May 2025*
  - Coursework: Probability and Statistics, Machine Learning, NLP, Data Structures, Database Systems

Technical Skills

- **Languages**: Python, C++, Java, SQL, R, JavaScript
- **ML & Deep Learning**: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers
- **LLM & Applied AI**: RAG, agentic pipelines, prompt engineering, embeddings & vector search, OpenAI API, MongoDB Atlas Vector Search
- **Backend & DevOps**: Flask, REST APIs, React, Docker, Auth0, Git
- **Data & Compute**: PostgreSQL, MySQL, MongoDB, Spark, Hadoop, SLURM/HPC, multi-GPU job scheduling
- **Cloud**: AWS (SageMaker, Lambda, EC2, S3, Redshift, Elastic Beanstalk, CloudFront), Snowflake, BigQuery
- **Certifications**: AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

Research Experience

- **Cross-Species Drug Translation using Optimal Transport & Generative AI** — Boston, MA
  Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute — *February 2026 – Present*
  - Built **Optimal Transport** models (**VAEs**, **unbalanced OT**) predicting animal-to-human trial translation from **scRNA-seq** atlases, with **scanpy** pipelines automating cell-type labeling and dataset integration.
  - Ported a 2018 CPU-only **TensorFlow** reference implementation to **PyTorch**, reimplementing the **Input Convex Neural Networks** behind the optimal-transport dual and enabling GPU training on Harvard's **FASRC Cannon** cluster.
  - Benchmarked **CellOT** against **scGen** on cross-species perturbation prediction, measuring 0.85–0.90 R^2 with the source species in training versus 0.65–0.67 on the held-out-species split.
  - Established through **paired single-factor ablations** that gene-space choice reverses with split — 6,619 orthologs wins in-distribution while 1,000 HVGs wins on every out-of-distribution metric — ruling out a split-independent default.
- **AlphaFold Protein–Nucleic Acid Interaction Prediction** [GitHub] — Atlanta, GA
  Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry — *February 2025 – August 2025*
  - Engineered a pipeline integrating **AlphaFold-Multimer** and **AlphaFold3** to predict protein–nucleic acid interactions from **MSA** data.
  - Automated high-throughput inference across GPU nodes with **SLURM** parallel scheduling, tuning batch size to raise GPU utilization on shared **HPC** clusters.
  - Built **Python** tooling (PyMOL, ChimeraX) ranking structures by **pLDDT** and **PAE** score, shipped as a documented CLI so non-technical researchers could run inference and triage results independently.
- **Computational Legislative Studies: UK & New Zealand Parliamentary Data** [GitHub] — Atlanta, GA
  Research Assistant — Emory University (Carrubba & Estrada) — *December 2023 – May 2024*
  - Built **Python** scrapers and **ETL** pipelines over UK Parliament open data, normalizing bills, members, divisions, and Hansard proceedings into a relational schema covering 22,606 member-level division votes, 4,791 MPs, 4,108 bills, and 3,875 constituencies.
  - Diagnosed **Cloudflare** bot mitigation (403 challenge responses) and restored collection using TLS-impersonating requests (`curl_cffi`) and **Selenium** against an otherwise blocked source.
  - Ran long scrapes as batch jobs on Emory's **QTM cluster** with isolated conda environments, staging data through **AWS S3/EC2** and documenting the pipeline for a UK-to-New-Zealand handoff.

ENGINEERING PROJECTS

- **Autonomous Ablation-Search Agent for Cluster-Scale ML Experimentation** [GitHub] Boston, MA
  Independent Project — Harvard FASRC Cannon Cluster *June 2026 – Present*
  - Built a **self-directed experiment harness** (submit → watch → reflect → decide → synthesize) that plans, launches, and triages **SLURM** ablation studies unattended — **338 runs**, 318 clean and 20 with handled errors, over **79 hours** of compute.
  - Split the planner from the execution substrate and **checkpointed each agenda to disk**, so the cluster keeps running the last plan while the planner is offline; an adaptive one-delta proposer clears a **validation gate** before every submit.
  - Designed a **fairshare-aware scheduler** routing small-MLP training to CPU partitions, where GPU carries 200–500× the fairshare cost, and made runs **preemption-tolerant** on `serial_requeue` via checkpoint/resume and capped job arrays.
  - Wired an optional **LLM planner** behind that interface with per-call **token and cost accounting** and a spend ceiling that degrades to the deterministic planner on breach or API failure.
- **Airway Management Simulation Chatbot** [Live Demo] Atlanta, GA
  Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital *January 2024 – August 2025*
  - Architected a clinical **agentic LLM pipeline** pairing **RAG** retrieval over **MongoDB Atlas Vector Search** with **prompt engineering** and context-window management to generate multi-turn training scenarios from **OpenAI** models.
  - Built a **HIPAA-conscious** full-stack platform (**Python/Flask** REST API, React, MongoDB, **Auth0**) deployed on **AWS** Elastic Beanstalk and CloudFront, tracking resident performance metrics.
  - Led an Agile team of 6, running code reviews and folding clinician feedback into successive releases.
- **Reimplementation of Transformer Architecture** [GitHub] Atlanta, GA
  Machine Learning Researcher — Emory Dept. of Computer Science *October 2024 – January 2025*
  - Implemented a Transformer in **PyTorch** from scratch — **Multi-Head Self-Attention**, **Positional Encodings**, **Layer Normalization** — validating architectural correctness against `nn.Transformer`.
  - Tuned hyperparameters and the **Cross-Entropy** objective to reach a 9.38 BLEU score on English-to-German translation.

MACHINE LEARNING COURSEWORK PROJECTS

- **Parameter-Efficient Fine-Tuning of BLIP-2 for Visual Question Answering** [GitHub] Boston, MA
  Harvard SHBT-261 *Spring 2026*
  - Fine-tuned **BLIP-2** (OPT-2.7B) on **TextVQA** using **LoRA**, lifting accuracy from 0.078 zero-shot to 0.213 — nearly 3× — while training 2.62M parameters (0.07% of the model) in 8 minutes on a single **A100**.
  - Isolated the contribution of **prompt engineering** across four templates, showing OCR-augmented prompting alone raised zero-shot accuracy to 0.106 before any weight updates.
  - Ablated **LoRA** rank over $r \in \{4, 8, 16\}$ and automated the pipeline end to end, generating every reported figure and LaTeX table directly from result JSONs for reproducibility.
- **Image Classification Benchmark: Classical ML, CNNs, and Vision Transformers** [GitHub] Boston, MA
  Harvard SHBT-261 *Spring 2026*
  - Benchmarked **ViT-B/16**, **ResNet-18**, and a HOG + Random Forest baseline on **Caltech-101** (8,677 images, 101 classes), reaching 0.972 top-1 and 0.998 top-5 with ViT against 0.507 for the classical pipeline.
  - Quantified the cost of that gain: ViT beat fine-tuned ResNet-18 by 1.8 points at roughly 9× the training time, making the accuracy-per-GPU-hour tradeoff explicit.
  - Established through ablations that light augmentation and **SGD** outperform heavy regularization and adaptive optimizers when fine-tuning on small datasets.
- **Semantic Segmentation Benchmark: U-Net, ViT, and DeepLabV3+** [GitHub] Boston, MA
  Harvard SHBT-261 *Spring 2026*
  - Benchmarked **DeepLabV3+**, **U-Net**, and a frozen **ViT-Tiny** encoder for semantic segmentation on **Pascal VOC 2007** (21 classes), reaching 0.640 mIoU and 0.928 pixel accuracy with COCO-pretrained DeepLabV3+.
  - Diagnosed **mode collapse** in U-Net under data scarcity — 209 training images yielded predictions for only 4 of 21 classes — isolating pretraining rather than architecture as the dominant factor.
  - Reached 0.304 mIoU in 105 s by training only an 814K-parameter decoder head on a frozen encoder, establishing a compute-efficient middle ground.
- **Pneumonia Detection from Chest Radiographs** [GitHub] Boston, MA
  Harvard BST-261 (Kaggle competition) *Spring 2026*
  - Diagnosed a 14-point generalization gap on a **Kaggle** chest-radiograph benchmark — 99% validation against 85% leaderboard — and traced it to memorization of a 4,700-image training set.
  - Closed the gap to 96% with domain-aware augmentation (**Mixup** $\alpha = 0.2$, RandomResizedCrop, RandomErasing, affine jitter), excluding vertical flips that would render chest radiographs anatomically implausible.
  - Found that freezing **DenseNet-121**'s early layers outweighed architecture choice, beating ResNet-50, ConvNeXt-Tiny, EfficientNet-B0, and a three-model ensemble to reach a 0.959 public score.